

$$y'' = y^2$$

a difficult differential equation

Try and solve

$$y'' = y^2$$

First:

$$\bullet y' = y^2$$

$$y = \frac{1}{A-x}$$

$$\bullet y'' = y$$

$$y = Ae^{-x} + Be^x$$

$$y' = y^2$$

$$\frac{dy}{dx} = y^2$$

$$\int \frac{1}{y^2} dy = \int dx$$

$$-\frac{1}{y} = x - A$$

$$y = \frac{1}{A-x}$$

$$y'' = y \Rightarrow y'' - y = 0$$

$$\left(\left(\frac{d}{dx} \right)^2 - I^2 \right) y = 0$$

$$\left(\frac{d}{dx} + I \right) \underbrace{\left(\frac{d}{dx} - I \right) y}_u = 0$$

$$\left(\frac{d}{dx} + I \right) \cdot u = 0 \Rightarrow u' = -u$$

$$\Rightarrow u = Ae^{-x} \Rightarrow y' - y = Ae^{-x} \%$$

$$\Rightarrow (y' - y = Ae^{-x}) \cdot e^{-x} \quad \text{integrating factor}$$

$$e^{-x} y' + \underbrace{(-e^{-x})}_{(e^{-x})'} y = Ae^{-2x}$$

$$\Rightarrow (e^{-x} y)' = Ae^{-2x} \quad \int \Rightarrow e^{-x} y = Ae^{-2x} + B$$

$$\Rightarrow y = Ae^{-x} + Be^x$$

$$\text{Now } y'' = y^2 \Rightarrow y'' \cdot y' = y^2 \cdot y'$$

$$\frac{1}{2} \left(\frac{d}{dx} (y')^2 \right) = \frac{1}{3} \frac{d}{dx} y^3$$

$$\frac{d}{dx} \left(\frac{1}{2} (y')^2 - \frac{1}{3} y^3 \right) = 0$$

$$\frac{1}{2} (y')^2 - \frac{1}{3} y^3 = A$$

$$(y')^2 = \frac{2}{3} y^3 + A$$

$$y' = \sqrt{\frac{2}{3}y^3 + A} \quad \rightarrow \quad \frac{dy}{dx} = \sqrt{\frac{2}{3}y^3 + A}$$

$$\int \frac{1}{\sqrt{\frac{2}{3}y^3 + A}} dy = \int dx = x + B$$

Elliptical Integral

Weierstrass - P - Function

$$y = 6^{\frac{1}{3}} P\left(\frac{x+C}{6^{\frac{1}{3}}}, 0, C_2\right)$$

Jump back to $y' = \sqrt{\frac{2}{3}y^3}$ with $A=0$ (simplification)

$$\frac{dy}{dx} = \sqrt{\frac{2}{3}} \cdot y^{3/2} \quad \Rightarrow \quad y^{-3/2} dy = \sqrt{\frac{2}{3}} dx$$

$$-2y^{-1/2} = \sqrt{\frac{2}{3}}x + B$$

$$\frac{1}{\sqrt{y}} = B - \frac{x}{\sqrt{6}} \quad \Rightarrow \quad y = \frac{1}{\left(B - \frac{x}{\sqrt{6}}\right)^2}$$

$$y = \frac{6}{(x-B)^2}$$